IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: **R26-SE-026**

1. Topic (12 words max)

A Unified Smart Platform for Roadside Assistance and Automotive Services in Sri Lanka

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Sri Lanka's transportation and automotive service ecosystem experiences significant inefficiencies during vehicle breakdowns, accidents, and routine maintenance. Roadside emergencies are typically managed through manual phone calls and fragmented service providers, resulting in delayed response times, increased driver stress, and prolonged road blockages. These challenges particularly affect daily commuters, commercial vehicle operators, elderly drivers, and women drivers, while also impacting traffic authorities and emergency responders.

The impact of this problem is multidimensional. Socially, delayed roadside assistance compromises road safety and heightens anxiety during emergencies. Economically, extended incident durations contribute to productivity losses, increased fuel consumption, and higher vehicle maintenance costs due to reactive servicing practices. Technically, the absence of an integrated digital platform limits real-time coordination, optimized dispatch, and data-driven decision-making. From a legal and ethical perspective, many existing solutions lack compliance with Sri Lanka's Personal Data Protection Act (PDPA), raising concerns regarding user privacy, consent, and data security. Environmentally, slow incident clearance increases traffic congestion, leading to unnecessary fuel wastage and increased emissions.

This research aligns with the SST – Software Systems & Technologies cluster and the Transportation / Smart Mobility industry vertical. It supports key United Nations Sustainable Development Goals, including SDG 9 (Industry, Innovation, and Infrastructure), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action). By proposing a unified, PDPA-compliant smart platform, this research addresses a critical national need and contributes to Sri Lanka's digital transformation in transport services.

6. Brief description of Existing Research and Systems (300 words max)

Modern roadside assistance platforms have transitioned from manual call-based services to mobile and web-based solutions that leverage GPS and real-time connectivity. Commercial systems such as **Honk** and **Urgently** provide on-demand roadside assistance by automatically matching stranded drivers with nearby service providers. These platforms offer services including towing, tire changes, fuel delivery, and lockout support, along with real-time tracking and estimated arrival times. While these systems demonstrate the effectiveness of digital dispatch and centralized service coordination, they are primarily designed for developed markets and operate within different regulatory and infrastructural contexts.

Academic research in this domain highlights the use of **location-based services, optimization algorithms, and intelligent dispatch systems** to reduce response times and improve operational efficiency. Studies show that automated dispatch and data-driven allocation significantly outperform manual coordination. Additionally, research in traffic management indicates that faster incident clearance can substantially reduce congestion and secondary delays. Predictive maintenance research further explores the use of IoT and machine learning to anticipate vehicle failures and shift maintenance from reactive to proactive models.

Despite these advancements, most existing systems focus on isolated functionalities such as dispatch or maintenance rather than a unified platform. There is limited consideration of **local constraints**, including older vehicle fleets, data availability, and regulatory requirements in developing countries like Sri Lanka. Moreover, compliance with data protection regulations such as Sri Lanka's **Personal Data Protection Act (PDPA)** is rarely addressed in existing solutions. These gaps highlight the need for a locally contextualized, integrated, and legally compliant smart platform.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed solution is a unified smart platform designed to integrate roadside assistance, automotive services, traffic intelligence, and predictive maintenance into a single ecosystem. The system directly addresses the fragmentation, inefficiency, and compliance gaps identified in existing solutions.

System Perspective (IPO Model)

Inputs:

1. User emergency requests with real-time location
2. Service provider availability (tow trucks, mechanics, garages)
3. Traffic and geospatial data
4. Vehicle usage and maintenance-related data
5. User consent and compliance parameters (PDPA)

Processes:

1. Smart dispatch optimization to assign the most suitable nearby service provider
2. Geo-intelligence analysis to assess traffic impact and prioritize incidents
3. Predictive maintenance models to anticipate vehicle failures
4. Secure data handling, consent management, and stakeholder integration

Outputs:

- Reduced response times and optimized service allocation
- Traffic impact visualizations and congestion reduction insights
- Maintenance alerts and service recommendations
- A trusted, PDPA-compliant unified user experience

The platform follows a modular software architecture, enabling scalability and future integration with government and insurance systems. A conceptual diagram (to be included separately) illustrates interactions between users, service providers, backend services, analytics modules, and compliance layers.

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

This project is supported by sufficient availability of specialized domain expertise, technical knowledge, and accessible data sources. The required resources are outlined below according to each core research component.

1. Dispatch Optimization & Smart Recovery

Expertise & Knowledge: The project team possesses foundational expertise in software engineering, algorithms, and system design, enabling the development of automated dispatch mechanisms. Established research in transportation optimization, vehicle routing problems (VRP), and emergency service allocation provides well-documented models and evaluation techniques. Academic literature and open-source algorithmic resources support the design and comparison of dispatch strategies.

Data Availability: While real-time roadside incident data is limited, dispatch scenarios can be simulated using synthetic datasets based on realistic assumptions. Public map APIs provide distance and travel-time estimates suitable for dispatch testing. Simulation-based evaluation is widely accepted in related academic studies.

2. Geo-Intelligence & Traffic Impact Analysis

Expertise & Knowledge: The project leverages knowledge in geospatial analysis, traffic modeling, and data visualization, supported by existing research on incident-induced congestion. GIS techniques and traffic simulation methodologies are well established in transportation research and are accessible through academic resources and open-source tools.

Data Availability: Traffic data can be obtained from publicly available sources such as map-based traffic layers and open traffic datasets. Where direct data access is limited, traffic simulation tools allow controlled experimentation to analyze congestion under delayed and rapid incident recovery scenarios. Visualization tools enable effective representation of results.

3. Predictive Maintenance for Service Management

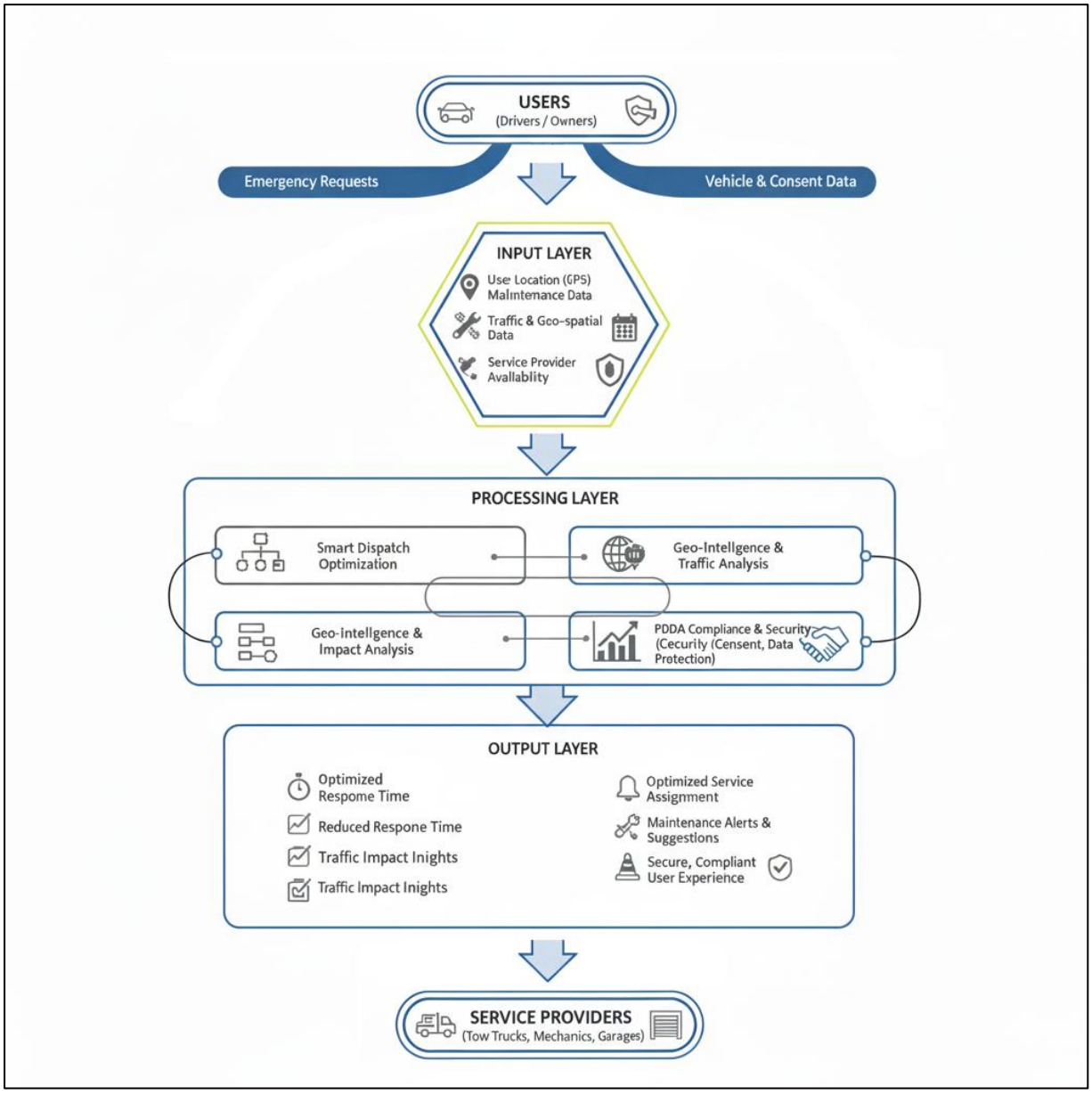
Expertise & Knowledge: Predictive maintenance knowledge is supported by extensive research in AI, machine learning, and IoT-based vehicle diagnostics. The team has the technical background to implement basic predictive models using standard machine learning frameworks. Literature provides guidance on selecting appropriate vehicle health indicators and evaluation metrics.

Data Availability: Although large-scale local vehicle sensor datasets are scarce, publicly available automotive datasets and synthetic data generation methods can be used. Standards such as OBD-II diagnostics offer structured data formats that can be simulated or collected in limited scope for research purposes.

4. Compliance, Integration & Stakeholder Engagement

Expertise & Knowledge: Regulatory expertise is supported by Sri Lanka's Personal Data Protection Act (PDPA) and official guidelines issued by the Data Protection Authority of Sri Lanka. Best practices from GDPR-compliant mobility platforms provide additional design references for privacy-by-design and data governance.

Data Availability: Legal and regulatory documentation is publicly accessible. Stakeholder knowledge can be gathered through surveys, interviews, and informal consultations with drivers, garages, and service providers. These qualitative inputs supplement technical design decisions without requiring restricted datasets.


USERS
(Drivers / Owners)
Emergency Requests
Vehicle & Consent Data
INPUT LAYER
Use: Location (GPS)
Malintenance Data
Traffic & Geo-spatial Data
Service Provider Availablity
PROCESSING LAYER
Smart Dispatch Optimization
Geo-Intellgence & Traffic Analysis
Geo-Intellgence & Impact Analysis
PDDA Compliance & Security (Cecurity (Censent, Data Protection)
OUTPUT LAYER
Optimized Respome Time
Reduced Respone Time
Traffic Impact Inights
Traffic Impact Inights
Optimized Service Assignment
Maintenance Alerts & Suggestions
Secure, Compliant User Experience
SERVICE PROVIDERS
(Tow Trucks, Mechanics, Garages)

9. Objectives and Novelty

Main Objective

To design and develop a unified, smart, and PDPA-compliant digital platform that integrates roadside assistance and automotive services in Sri Lanka, enabling real-time service requests, optimized provider dispatch, and improved coordination to reduce response times, traffic congestion, and user stress during vehicle-related incidents.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Janukshan S IT22635266	To develop and evaluate a smart dispatch mechanism that minimizes roadside assistance response time through automated service provider allocation.	• Analyze current manual dispatch limitations • Design proximity-based and rule-based dispatch logic • Simulate and compare dispatch strategies • Measure response time and efficiency improvements	Introduces algorithm-driven dispatch optimization to Sri Lanka's largely manual roadside assistance system, a concept not addressed in local research.
Asath M M IT22633422	To quantify the impact of incident response time on traffic congestion using geo-spatial analysis.	• Collect or simulate traffic and incident data • Analyze congestion under delayed vs rapid recovery • Visualize traffic impact using maps and heatmaps	Links roadside recovery speed with traffic congestion reduction in the Sri Lankan context, where such analysis is currently lacking.

Herath D M S T IT22639776	To explore predictive maintenance techniques to reduce unplanned vehicle breakdowns.	• Review AI/IoT-based maintenance models • Develop or simulate failure prediction methods • Integrate predictive alerts into the platform	Shifts Sri Lanka's maintenance approach from reactive to predictive, adapting advanced methods to limited-data environments.
De Silva R K D H IT22001252	To ensure PDPA-compliant data handling and integrate multiple automotive stakeholders into a unified platform.	• Design consent and data protection mechanisms • Plan integration with service providers and authorities • Define governance and access control policies - Gather stakeholder feedback and validate the platform against PDPA requirements	Embeds PDPA compliance and ecosystem integration into platform design an aspect largely missing in existing mobility solutions.

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes		No	

b) The proposed topic exhibits novelty.

Yes		No	

c) The student group can successfully execute the proposed project.

Yes		No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes		No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				